

Linked Open Data & Programming

IE650 Knowledge Graphs



Overview

- Linked Open Data
 - Principles
 - Examples
 - Vocabularies
- Microdata & schema.org
- Introduction to Semantic Web Programming with rdflib & Jena

Linked Open Data

- What we've got to know up to now
 - RDF as a universal language for encoding knowledge
 - RDF Schema for describing vocabularies (i.e., classes and properties)
- How can we publish such knowledge?
- Linked Open Data
 - uses techniques like URIs, RDF, RDF schema
 - for publishing knowledge on the Web

Why “Linked” Open Data?

```
:p a :Physician .  
:p :hasDegree "Dr." .  
:p :hasName "Mark Smith" .  
:p :hasAddress :a .  
:a :street "Main Street" .  
:a :number "14"^^xsd:int .  
:a :city "Smalltown" .  
:p :hasOpeningHours [  
  a rdf:Bag ;  
  [ :day :Monday;  
    :from "9"^^xsd:int;  
    :to "11"^^xsd:int;  
  ]  
  ...  
]
```



```
:s a :City .  
:s :name "Smalltown" .  
:s :lat "49.86"^^xsd:double .  
:s :long "8.65"^^xsd:double .  
:s :district "Birmingham" .  
...
```



```
:d a :District .  
:d :name "Birmingham" .  
:d :pop "347891"^^xsd:int .  
:d :locatedIn "England" .  
...
```



Why “Linked” Open Data?

- Information is scattered on the Web
 - Publishing your own knowledge graph online just adds a scattered piece
 - “information silos”
- HTML has a concept for interlinking scattered information
 - known as *hyperlink*
 - More information at `<a href="http://www.w3.org">W3C</a>`
- Linked Open Data uses that principle, too

Why “Linked” Open Data?

```
:p a :Physician .  
:p :hasDegree "Dr." .  
:p :hasName "Mark Smith" .  
:p :hasAddress :a .  
:a :street "Main Street" .  
:a :number "14"^^xsd:int .  
:a :city  
  <http://.../smalltown>  
:p :hasOpeningHours [  
  a rdf:Bag ;  
  [ :day :Monday;  
    :from "9"^^xsd:int;  
    :to "11"^^xsd:int;  
  ]  
] ...
```

```
:s a :City .  
:s :name "Smalltown" .  
:s :lat "49.86"^^xsd:double .  
:s :long "8.65"^^xsd:double .  
:s :district  
  <http://.../birmingham> .  
...
```

```
:d a :District .  
:d :name "Birmingham" .  
:d :pop "347891"^^xsd:int .  
:d :locatedIn "England" .  
...
```



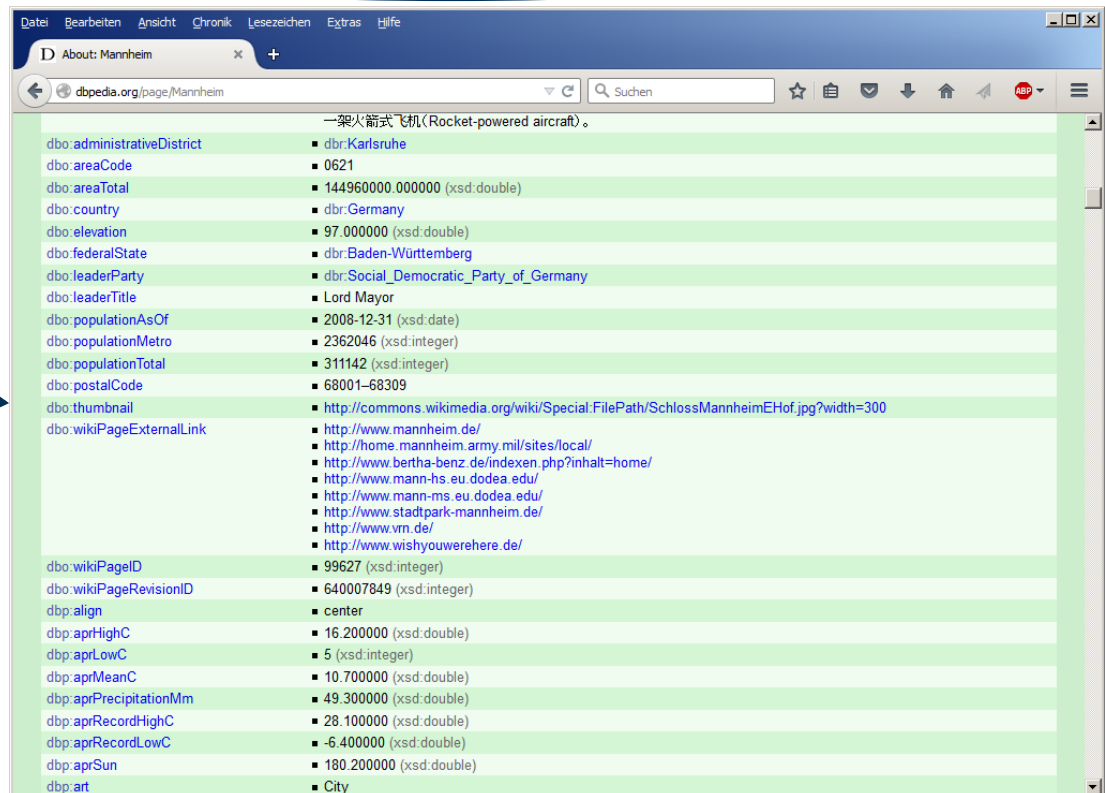
Why “Linked” Open Data?

- Linked Open Data is RDF data
 - Which is provided in a distributed manner
- URIs
 - have been used as simple identifiers so far
 - in LOD: links to data
 - Resolvable!
 - "dereferencable URIs" (URLs)
 - Can be used together with content negotiation, RDFa, etc.

Why “Linked” Open Data?

- Example:

<#Heiko> :worksIn <<http://dbpedia.org/resource/Mannheim>>.

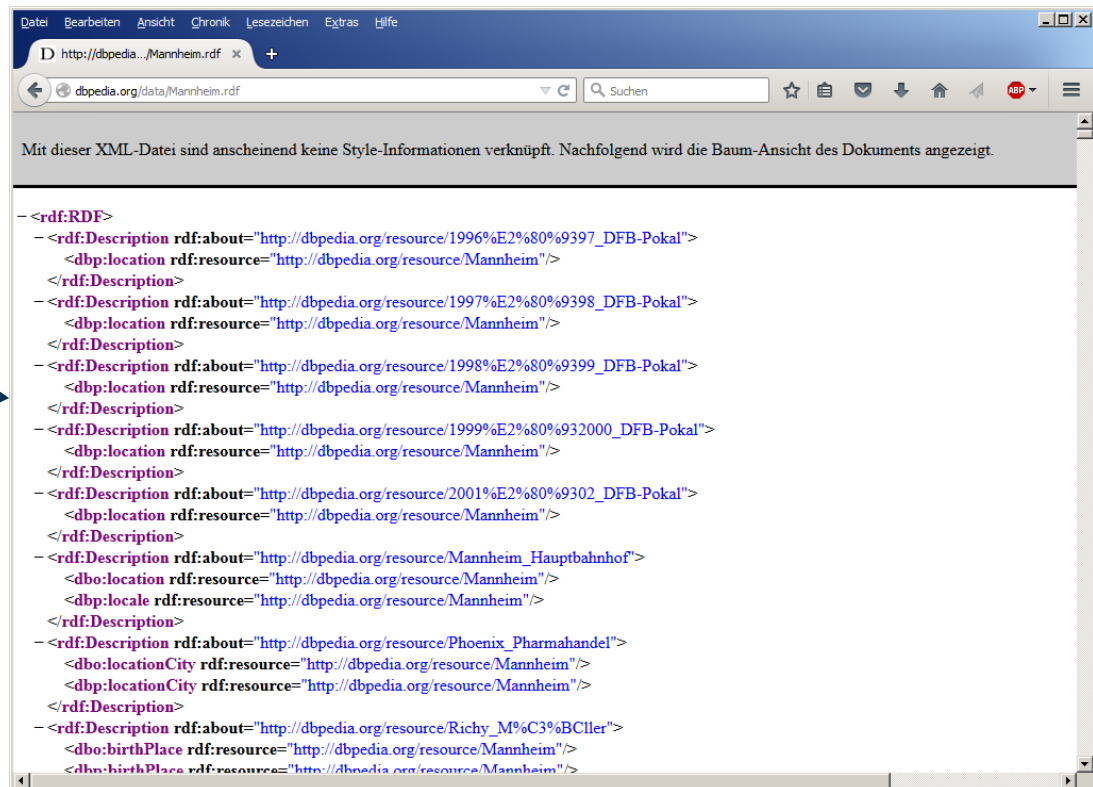


dbo:administrativeDistrict	■ dbr:Karlsruhe
dbo:areaCode	■ 0621
dbo:areaTotal	■ 144960000.000000 (xsd:double)
dbo:country	■ dbr:Germany
dbo:elevation	■ 97.000000 (xsd:double)
dbo:federalState	■ dbr:Baden-Württemberg
dbo:leaderParty	■ dbr:Social_Democratic_Party_of_Germany
dbo:leaderTitle	■ Lord Mayor
dbo:populationAsOf	■ 2008-12-31 (xsd:date)
dbo:populationMetro	■ 2362046 (xsd:integer)
dbo:populationTotal	■ 311142 (xsd:integer)
dbo:postalCode	■ 68001–68309
dbo:thumbnail	■ http://commons.wikimedia.org/wiki/Special:FilePath/SchlossMannheimEHof.jpg?width=300
dbo:wikiPageExternalLink	■ http://www.mannheim.de/ ■ http://home.mannheim.army.mil/sites/local/ ■ http://www.bertha-benz.de/indexen.php?inhalt=home/ ■ http://www.mann-hs.eu.dodea.edu/ ■ http://www.mann-ms.eu.dodea.edu/ ■ http://www.stadtpark-mannheim.de/ ■ http://www.vm.de/ ■ http://www.wishyouwerehere.de/
dbo:wikiPageID	■ 99627 (xsd:integer)
dbo:wikiPageRevisionID	■ 640007849 (xsd:integer)
dbp:align	■ center
dbp:aprHighC	■ 16.200000 (xsd:double)
dbp:aprLowC	■ 5 (xsd:integer)
dbp:aprMeanC	■ 10.700000 (xsd:double)
dbp:aprPrecipitationMm	■ 49.300000 (xsd:double)
dbp:aprRecordHighC	■ 28.100000 (xsd:double)
dbp:aprRecordLowC	■ -6.400000 (xsd:double)
dbp:aprSun	■ 180.200000 (xsd:double)
dbp:art	■ City

Why “Linked” Open Data?

- Example:

<#Heiko> :worksIn <http://dbpedia.org/resource/Mannheim>.



HTML Links vs. Links in Linked Open Data

- Compare

Heiko works in `<a href="http://www.mannheim.de">Mannheim</a>.`

to

`:Heiko :worksIn <http://dbpedia.org/resource/Mannheim> .`

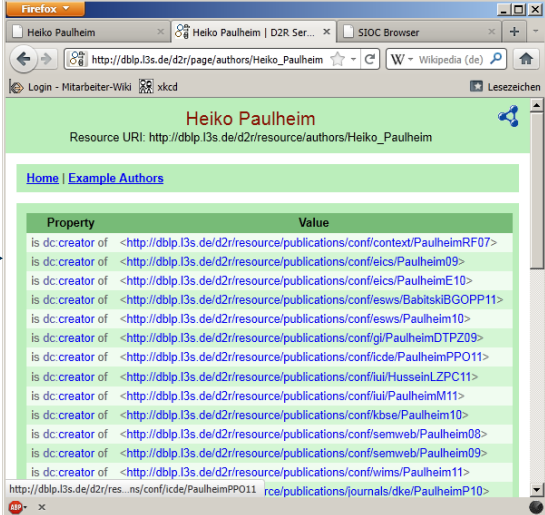
- Observation:

- Links in Linked Open Data are always *explicitly* typed
- The semantics of the link is thus interpretable
 - Given that the predicate is defined in a schema

Links in Linked Open Data

- Important special case: owl:sameAs*

```
:Heiko  
  owl:sameAs  
    <http://dblp.13s.de/d2r/page/  
      authors/Heiko_Paulheim>
```



Property	Value
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/context/PaulheimRF07>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/eics/Paulheim09>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/eics/PaulheimE10>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/esws/BabitskiBGOPP11>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/esws/Paulheim10>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/gi/PaulheimDTPZ09>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/icde/PaulheimPPO11>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/ii/HusseinLZPC11>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/iii/PaulheimM11>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/kbse/Paulheim10>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/semweb/Paulheim08>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/semweb/Paulheim09>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/conf/vims/Paulheim11>
is dc: creator of	<http://dblp.13s.de/d2r/resource/publications/journals/dke/PaulheimP10>



* We don't know OWL yet, never mind, we'll get to that...

Links in Linked Open Data

- Important special case: owl:sameAs*
- Links two *identical* resources
 - This is required due to the non-unique naming assumption
- One of the most commonly misused concepts in the Semantic Web...
- Use:
 - Two datasets with information about the same person
- Abuse:
 - A dataset with information about a person and the person's homepage
 - The Starbucks in O7 and the company Starbucks
 - The state and the city of Hamburg
 - The parliament as an institution and the parliament as a building

* We don't know OWL yet, never mind, we'll get to that...

Links in Linked Open Data

- Alternatives to abusing `owl:sameAs`*
 - General link to other resources
`rdfs:seeAlso`
 - Link to (HTML) homepage:
e.g., `foaf:homepage`

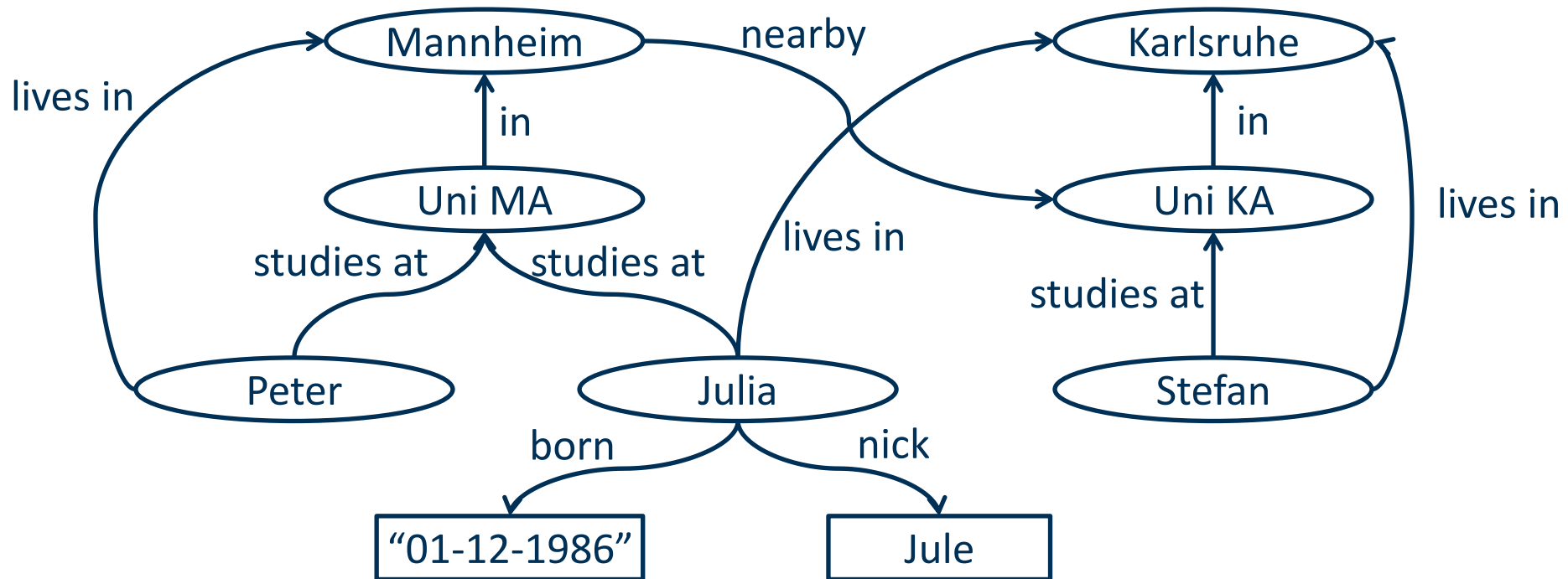
Four Principles of Linked Open Data

- The four Principles by Tim Berners-Lee (2006)
 - 1) Use URIs to identify things
 - 2) Use derefencable URIs
 - 3) Provide useful information upon derefencable URIs, use standards
 - 4) Add links to other datasets



RDF Molecules: Theory and Practice

- Example Graph



The Five Star Schema

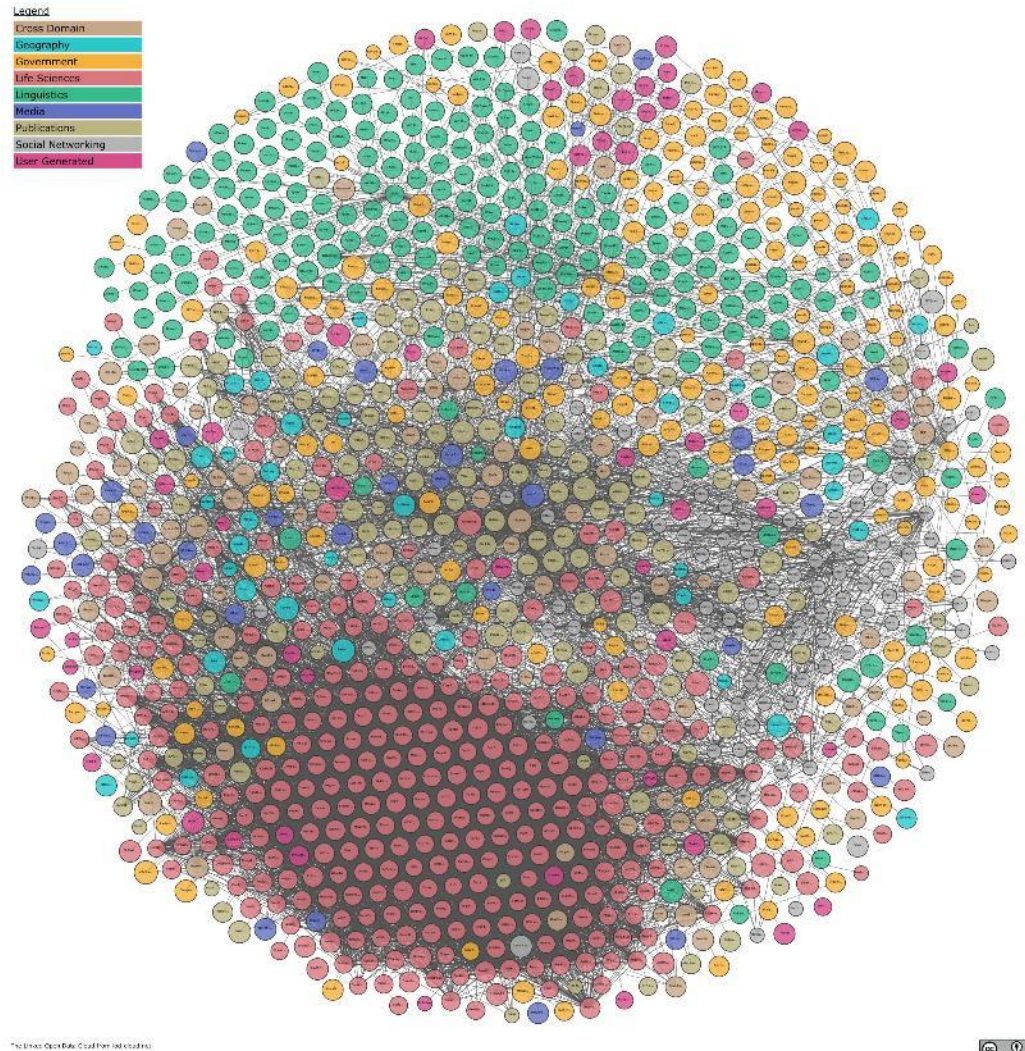
- Five Star Scheme (Tim Berners-Lee, 2010)
 - * Available on the web with an open license
 - ** Available as machine-readable, structured data
 - *** like ** plus using a non-proprietary format
 - **** like*** plus using open standards by the W3C
 - ***** like **** plus links to other datasets



Linked Open Data Best Practices

- as defined by Heath and Bizer, 2011
 - 1) Provide dereferencable URIs
 - 2) Set RDF links pointing at other data sources
 - 3) Use terms from widely deployed vocabularies
 - 4) Make proprietary vocabulary terms dereferencable
 - 5) Map proprietary vocabulary terms to other vocabularies
 - 6) Provide provenance metadata
 - 7) Provide licensing metadata
 - 8) Provide data-set-level metadata
 - 9) Refer to additional access methods

The Linked Open Data Cloud

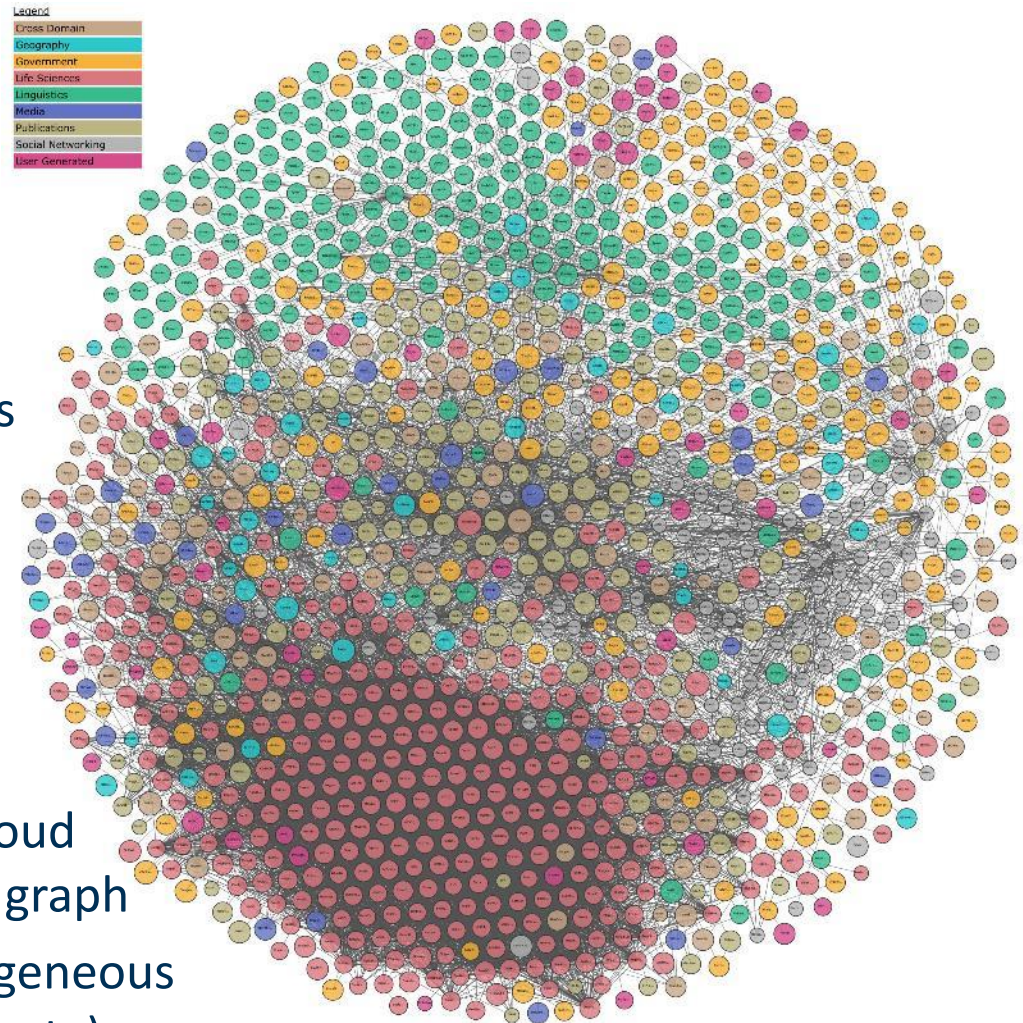


The Linked Open Data Cloud (lod-cloud.net)



What is the Linked Open Data Cloud?

- Viewpoint 1: a set of interconnected knowledge graphs
 - People have published ~1,000 knowledge graphs
 - They are linked to one another
- Viewpoint 2: one huge knowledge graph
 - In its entirety, the LOD cloud forms a large knowledge graph
 - This graph is very heterogeneous (i.e., uses different schemata)

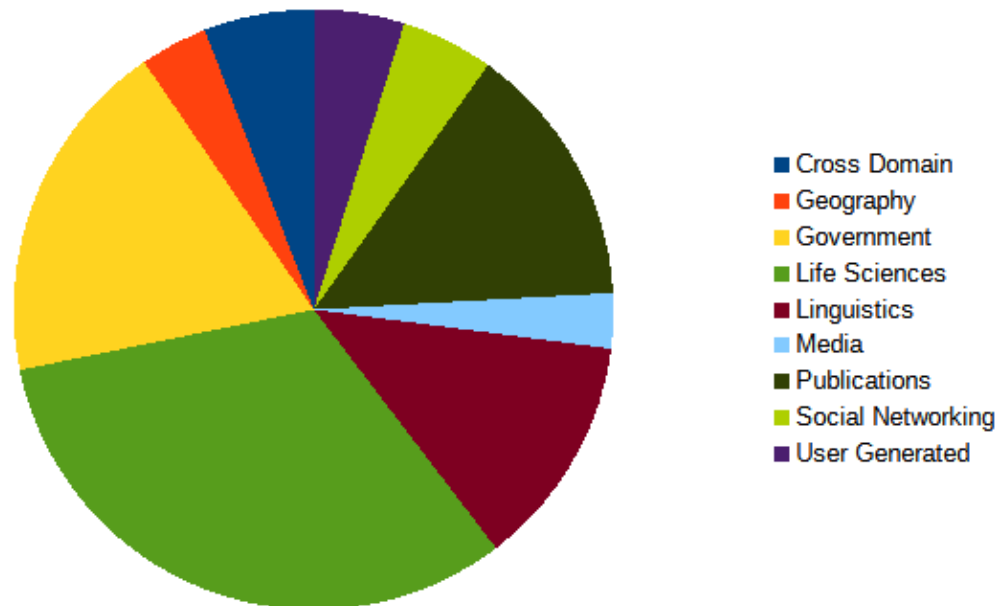


The Linked Open Data Cloud

- In numbers:
 - >1,250 Data sets
 - Several billion triples
 - Several million interlinks
- Topical domains:
 - Government
 - Publications
 - Life sciences
 - User-generated content
 - Cross-domain
 - Media
 - Geographic
 - Social web

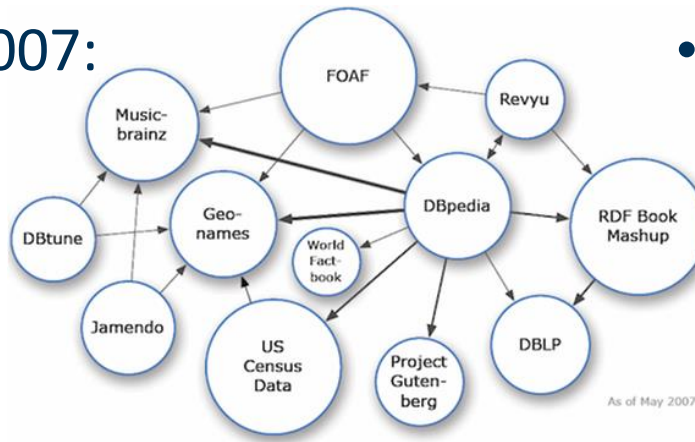
The Linked Open Data Cloud

- Domains by number of datasets in Linked Open Data
 - As of 2019
 - Classified based on data provider tags
 - More than half of the datasets are government and life sciences

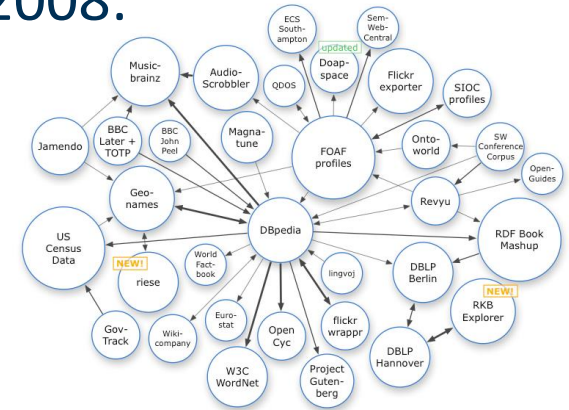


A Short History of Linked Open Data

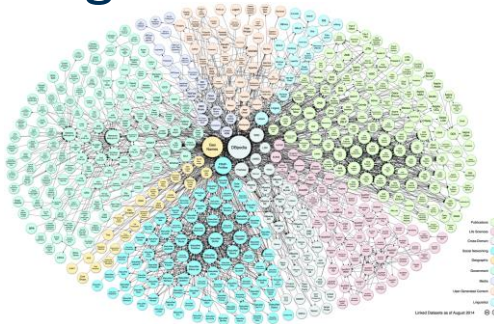
- May 2007:



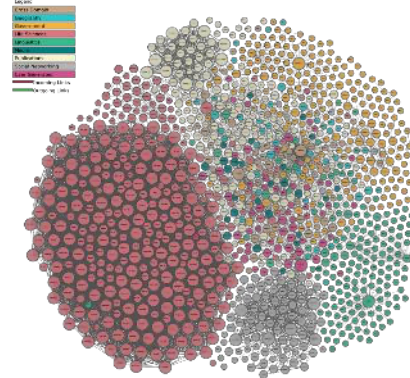
- March 2008:



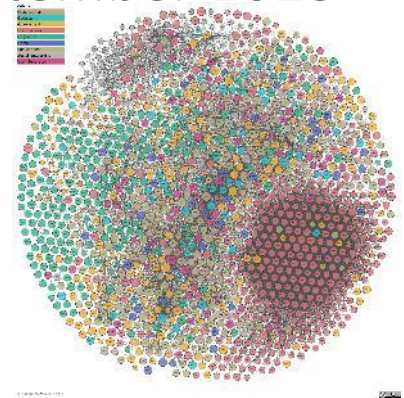
- August 2014



- August 2017



- September 2023



Examples: Government Data

 DATA.GOV

DATA TOPICS - IMPACT APPLICATIONS DEVELOPERS CONTACT

DATA CATALOG / Datasets Organizations ?

Search datasets...

Order by: Popular

Datasets ordered by Popular

Formats: RDF

Filter by location: Enter location...

11,717 datasets found

Demographic Statistics By Zip Code 1186 recent views
City of New York — Demographic statistics broken down by zip code
CSV RDF JSON XML

Popular Baby Names 213 recent views
City of New York — Popular Baby Names by Sex and Ethnic Group Data were collected through civil birth registration. Each record represents a baby born in New York City.
CSV RDF JSON XML

Accidental Drug Related Deaths
State of Connecticut — A listing of deaths from 2012 to 2018. A "Y" value under the "Cause of Death" column indicates a death due to an accidental drug-related death.
CSV RDF JSON XML

SAT School Participation and Performance
State of Connecticut — This dataset contains data on student participation and performance on the State of Connecticut College and Career Readiness (CCR) assessment.
CSV RDF JSON XML

Map tiles & Data by OpenStreetMap, under CC BY-SA

Topics: Local Government (9938) Climate (5) Disasters (3)

Topic Categories: Clear All

data.gov.uk | Find open data Publish your data Documentation Support

We've been improving data.gov.uk to help you find and use open government data. Discover what's changed and get in touch to give us your feedback.

[Don't show this message again](#)

Search results

Search bar with magnifying glass icon

Filter by 285 results found Best match ▼

Sitemap Legal notice Contact English (en)

Roles & Salaries

us Fraud Office
October 2016
(part) showing all staff roles. Names and
Senior Civil Servants. Organogram data is
ment departments and...

Roles & Salaries

EU Open Data Portal

Access to European Union open data

EUROPA > EU Open Data Portal > Linked Data

Home Data Applications **Linked data** Visualisations Developers' corner About

About linked data

Linked data is a standard way to represent data on a wide range of topics. Publishing linked data makes it easier for developers to connect information from different sources, resulting in new and innovative applications.

Need help?

Any questions or queries about Linked Data?

Contact us

SPARQL

You can search for the metadata stored in the EU Open Data Portal triple store by using the SPARQL endpoint query editor below.

Namespaces

PREFIX dcat: <http://www.w3.org/ns/dcat#>
PREFIX odp: <http://data.europa.eu/euodp/ontologies/ec-odp#>
PREFIX dc: <http://purl.org/dc/terms/>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX foaf: <http://xmlns.com/foaf/0.1/>

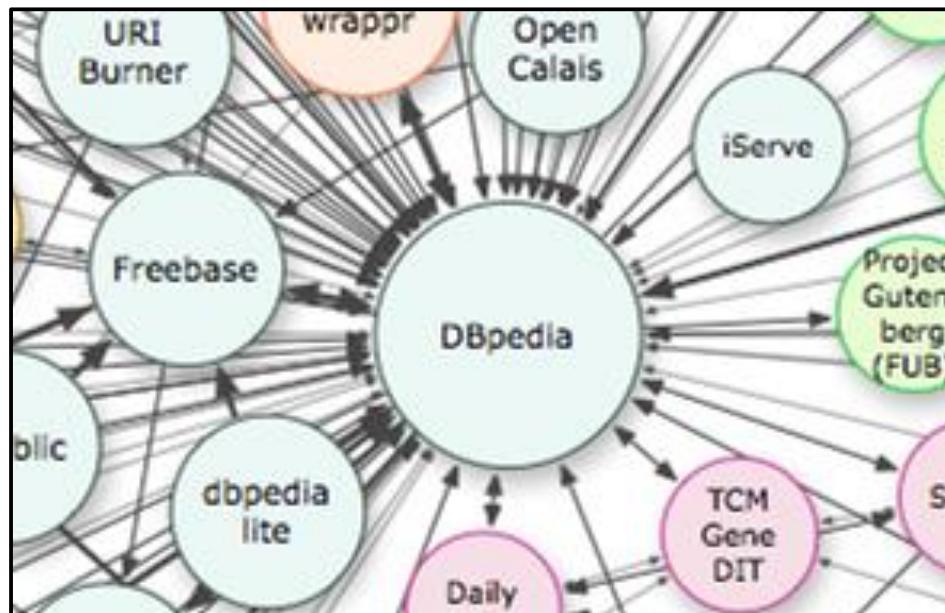
Sample queries

For the EU Open Data Portal metadata catalogue

Retrieve dataset with specific title (eg. 'Register of Commission documents')
Retrieve number of datasets per publisher
Retrieve all publishers
Retrieve all datasets that have been modified after a certain date
Retrieve all the resources from a dataset with a title that contains specific words (eg. 'Register of Commission documents')

Cross-Domain Example: DBpedia

- General knowledge on almost five million entities
- Hundreds of millions of triples
- Linked to ~100 other datasets
 - The most interlinked dataset



DBpedia: How It Is built

University of Mannheim



Motto	<i>In Omnibus Veritas</i> <i>Suprema Lex Esto</i> (Latin)
Motto in English	Truth in everything should be the supreme law
Established	1763: Theodoro Palatinae 1907: Handelshochschule 1967: Universität Mannheim
Type	Public
Endowment	€115 million
Chancellor	Susann-Annette Storm
Rector	Ernst-Ludwig von Thadden
Academic staff	800 (full time)
Administrative staff	550 (full time)
Students	12,151 (HWS 2013/14) ^[1]
Undergraduates	6,915 ^[1]
Postgraduates	4,965 ^[1]
Doctoral students	249 ^[1]

```

{{Infobox university
|motto           = ''In Omnibus Veritas Suprema Lex Esto'' ([[Latin]])
|mottoeng        = Truth in everything should be the supreme law
|name            = University of Mannheim
|native_name     = Universität Mannheim
|image_name      = Uni_Mannheim_Siegel.gif
|caption         = [[Seal (emblem)|Seal]] of the UMA
|established     = 1763: Theodoro Palatinae <br/> 1907: Handelshochschu
|type            = [[Public University|Public]]
|endowment       = €115 [[million]]
|academic_staff  = 800 (full time)
|administrative_staff = 550 (full time)
|Schools         = 5
|rector          = [[Ernst-Ludwig von Thadden]]
|chancellor      = [[Susann-Annette Storm]]
|students        = 12,151 <small>''(HWS 2013/14)''</small><ref name="un
/Studierendenstatistik_hws13.pdf|title= Studierendestatistik der Univ
|undergrad       = 6,915<ref name="uni-mannheim.de"/>
|postgrad        = 4,965<ref name="uni-mannheim.de">
|doctoral        = 249<ref name="uni-mannheim.de">
|profess         = 
|city            = [[Mannheim]]
|state           = [[Baden-Württemberg]]
|country         = [[Germany]]
|coord           = {{Coord
<?xml version="1.0" encoding="utf-8" ?>
<rdf:RDF
  <rdf:Description rdf:about="http://dbpedia.org/resource/University_of_Mannheim">
    <dbp:mottoeng xml:lang="en">Truth in everything should be the supreme law</dbp:mottoeng>
    <dbp:chancellor xml:lang="en">Susann-Annette Storm</dbp:chancellor>
    <dbp:budget rdf:datatype="http://dbpedia.org/datatype/euro">1.23E8</dbp:budget>
    <dbp:campus xml:lang="en">Urban , 74 acres</dbp:campus>
    <dbp:caption xml:lang="en">Seal of the UMA</dbp:caption>
    <dbp:city rdf:resource="http://dbpedia.org/resource/Mannheim" />
    <dbp:colors xml:lang="en">Mannheim Blue and White</dbp:colors>
    <dbp:country rdf:resource="http://dbpedia.org/resource/Germany" />
    <dbp:imageName xml:lang="en">Uni_Mannheim_Seal.png</dbp:imageName>
    <dbp:logo xml:lang="en">University of Mannheim.svg</dbp:logo>
    <dbp:logoSize rdf:datatype="http://www.w3.org/2001/XMLSchema#integer">280</dbp:logoSize>
    <dbp:nativeName xml:lang="en">Universität Mannheim</dbp:nativeName>
    <dbp:rector xml:lang="en">Thomas Puhl</dbp:rector>
    <dbp:state rdf:resource="http://dbpedia.org/resource/Baden-Württemberg" />
    <dbp:type rdf:resource="http://dbpedia.org/resource/Public_University" />
    <dbp:website rdf:resource="https://www.uni-mannheim.de/en/%7C2=www.uni-mannheim.de" />
    <georss:point>49.4832 8.4647</georss:point>
  </rdf:Description>
</rdf:RDF>

```

DBpedia: Contents

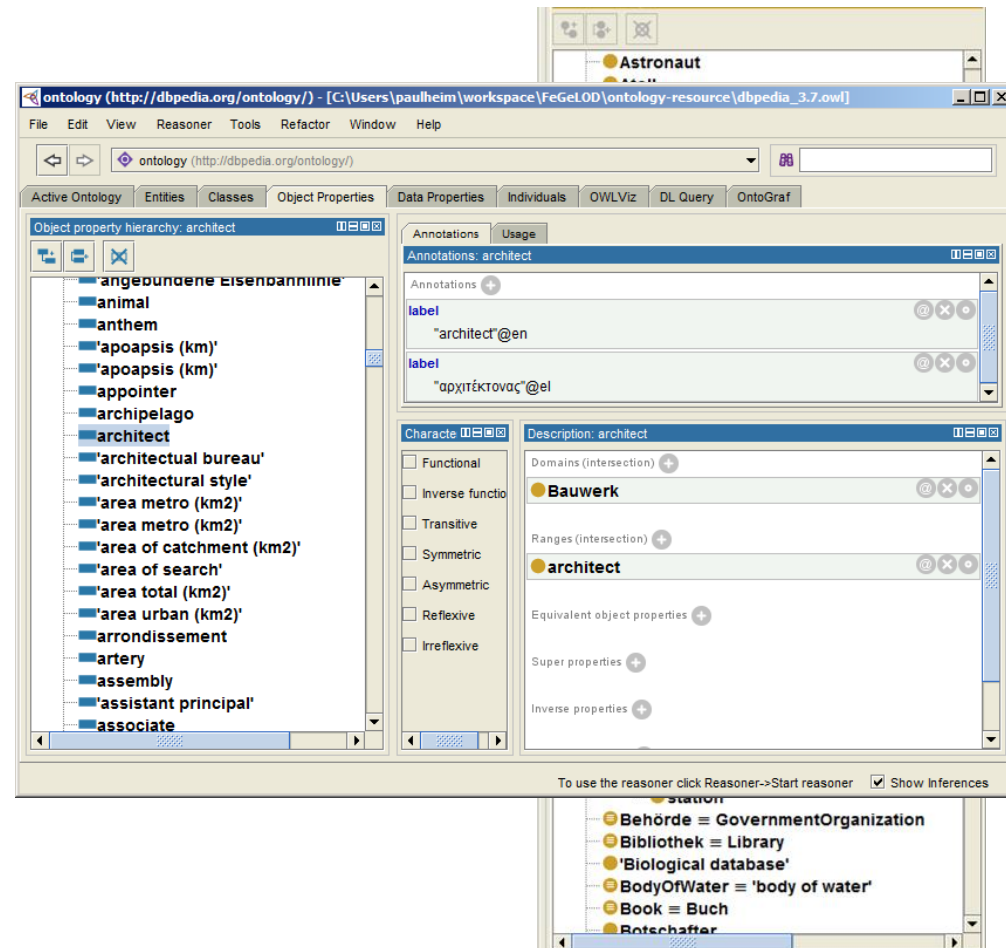
- Data from different infoboxes (extracted from multiple languages)
- Redirects and disambiguations
- External web links
- Abstracts in multiple languages
- Instance type information
 - DBpedia Ontology
 - YAGO*
 - schema.org*
 - DOLCE**
 - ...and others

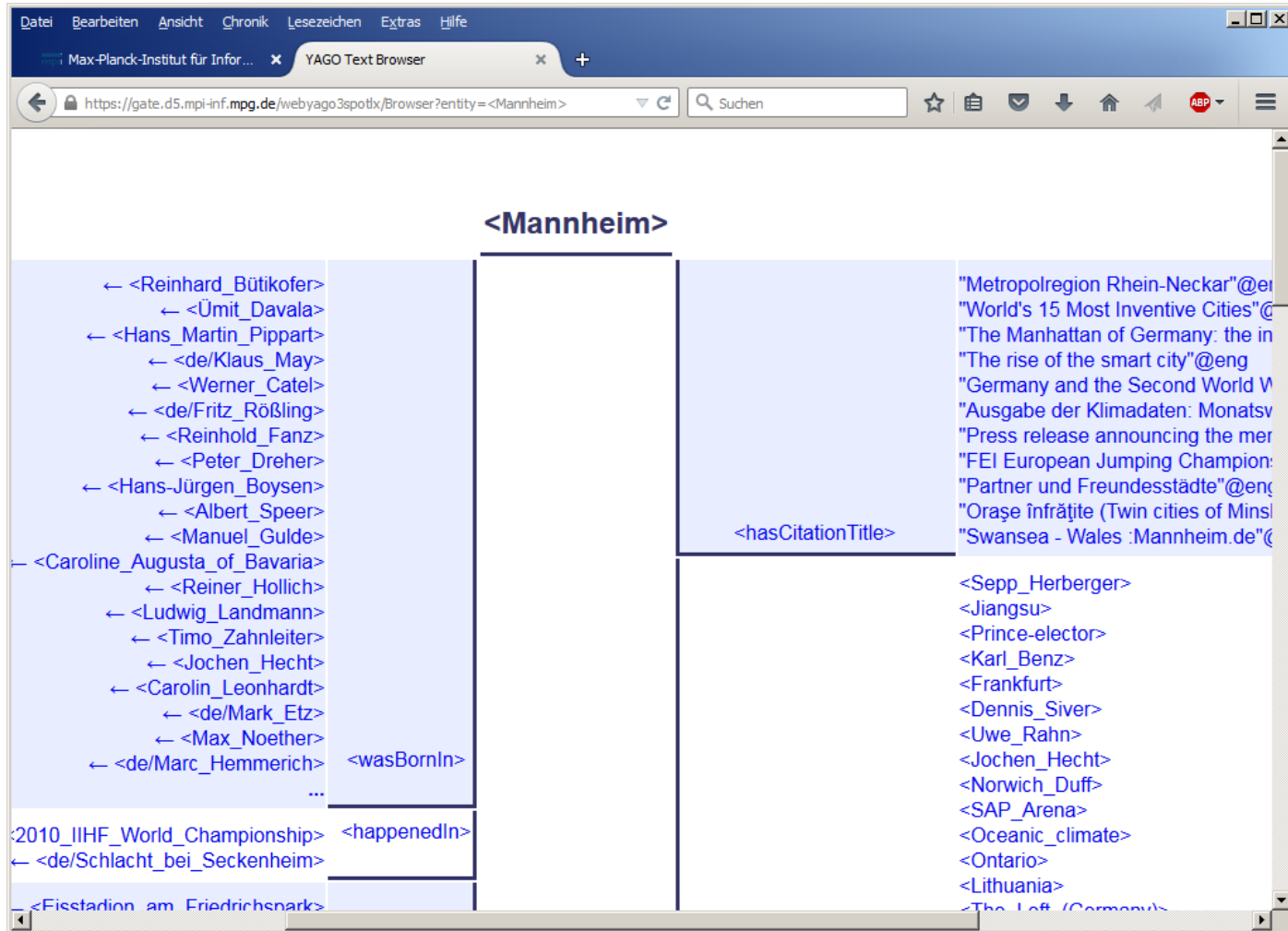
* later today

** in a few weeks

The DBpedia Ontology

- Classes:
 - ~1,800 classes
 - partial hierarchy
- Properties:
 - ~1,200 relations
 - many with domain/range
 - ~1,700 data properties
 - i.e., literal-valued
 - a bit of hierarchy





The screenshot shows the YAGO Text Browser interface. The browser window has a single tab titled "YAGO Text Browser" and the address bar shows the URL: `https://gate.d5.mpi-inf.mpg.de/web/yago3spotlx/Browser?entity=<Mannheim>`. The main content area displays a knowledge graph for the entity **<Mannheim>**.

The graph is structured as follows:

- Left Column (Entities):** A list of entities, each preceded by a left-pointing arrow (`<`). The entities include:
 - `<Reinhard_Bütikofer>`
 - `<Ümit_Davala>`
 - `<Hans_Martin_Pippart>`
 - `<de/Klaus_May>`
 - `<Werner_Catel>`
 - `<de/Fritz_Rößling>`
 - `<Reinhold_Fanz>`
 - `<Peter_Dreher>`
 - `<Hans-Jürgen_Boysen>`
 - `<Albert_Speer>`
 - `<Manuel_Gulde>`
 - `<Caroline_Augusta_of_Bavaria>`
 - `<Reiner_Hollich>`
 - `<Ludwig_Landmann>`
 - `<Timo_Zahnleiter>`
 - `<Jochen_Hecht>`
 - `<Carolin_Leonhardt>`
 - `<de/Mark_Etz>`
 - `<Max_Noether>`
 - `<de/Marc_Hemmerich>`
 - `<2010_IIHF_World_Championship>`
 - `<de/Schlacht_bei_Seckenheim>`
 - `<Eisstadion am Friedrichspark>`
- Right Column (Citations):** A list of citation titles, each preceded by a left-pointing arrow (`<`). The citations include:
 - `<Metropolregion Rhein-Neckar">`
 - `<"World's 15 Most Inventive Cities">`
 - `<"The Manhattan of Germany: the in">`
 - `<"The rise of the smart city">`
 - `<"Germany and the Second World W">`
 - `<"Ausgabe der Klimadaten: Monatsv">`
 - `<"Press release announcing the mer">`
 - `<"FEI European Jumping Champion:">`
 - `<"Partner und Freundesstädte">`
 - `<"Orașe înfrățite (Twin cities of Mins">`
 - `<"Swansea - Wales :Mannheim.de">`
 - `<Sepp_Herberger>`
 - `<Jiangsu>`
 - `<Prince-elector>`
 - `<Karl_Benz>`
 - `<Frankfurt>`
 - `<Dennis_Siver>`
 - `<Uwe_Rahn>`
 - `<Jochen_Hecht>`
 - `<Norwich_Duff>`
 - `<SAP_Arena>`
 - `<Oceanic_climate>`
 - `<Ontario>`
 - `<Lithuania>`
 - `<The_Left_(Germany)>`

The graph is organized into a table-like structure with three main sections: a list of entities on the left, a central column for the entity name, and a list of citations on the right. The entities are listed in a column, and the citations are listed in a column. The central column contains the entity name **<Mannheim>**.

- Also derived from Wikipedia
 - ~4.6M entities
 - ~26M statements
- Uses Wikipedia categories for typing
 - a class hierarchy of ~500,000 types
- Tries to capture time
 - i.e., statements that held true for a period of time
 - e.g., soccer players playing for teams
 - uses reification

Search: eng ▾

<id_1u5xrvs_1ul_zxcbb2>

<Miroslav_Klose> <playsFor> <FC_Bayern_Munich>	hasFactId	<extractionSource>	<http://en.wikipedia.org/wiki/Miroslav_Klose> → <http://en.wikipedia.org/wiki/Miroslav_Klose> →
		<occursUntil>	"2011-##-##"^^xsd:date →
		<occursSince>	"2007-##-##"^^xsd:date →

- Collaboratively edited knowledge base
- Size
 - ~15M instances
 - ~66M statements
- Ontology
 - ~23k classes
 - ~1.6k properties
- Special
 - provenance information
 - i.e., evidence: where did that statement come from?



- Main page
- Community portal
- Project chat
- Create a new item
- Item by title
- Recent changes
- Random item
- Help
- Donate

Print/export

- Create a book
- Download as PDF
- Printable version

Tools

- What links here
- Related changes
- Special pages
- Permanent link
- Page information
- Concept URI
- Cite this page

Item [Discussion](#)

Trent Reznor (Q282722)

American musician

No aliases defined

[► In more languages](#)

Statements

sex or gender

QID

male

[\[edit\]](#)

▼ 4 references

[\[edit\]](#)

imported from Swedish Wikipedia

[\[edit\]](#)

imported from Virtual International Authority File

[\[edit\]](#)

imported from Italian Wikipedia

[\[edit\]](#)

stated in Integrated Authority File

retrieved 27 April 2014

[\[add reference\]](#)

[\[edit\]](#)

Wikipedia (33 entries) [\[edit\]](#)

[\[Collapse\]](#)

be_x_old Трэнт Рэзнар

be Трэнт Рэзнар

bg Трент Резнър

cs Trent Reznor

da Trent Reznor

de Trent Reznor

en Trent Reznor

es Trent Reznor

et Trent Reznor

fa ترنت رزرنر

fi Trent Reznor

fr Trent Reznor

gl Trent Reznor

hu Trent Reznor

id Trent Reznor

is Trent Reznor

it Trent Reznor

ja トレント・レズナー

ka ტრენტ რეზნორი

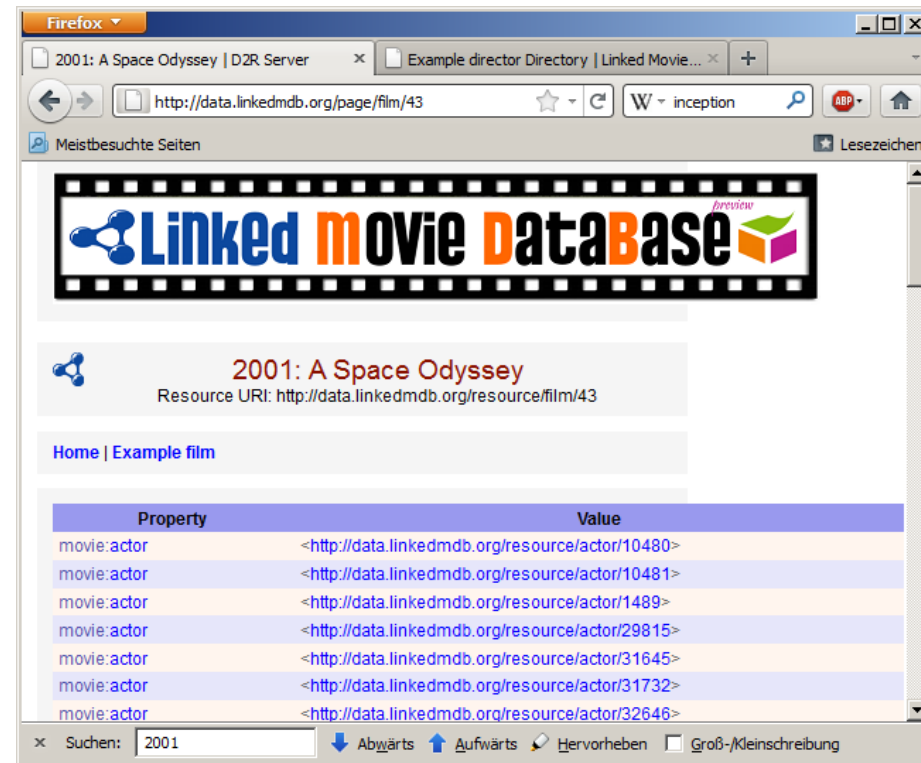
ko 트레트 레즈너

lv Tvents Reznors

nl Trent Reznor

Further Example Datasets

- Linked Movie Database
 - Movies, actors, directors...
- MusicBrainz
 - Artists, albums, ...
- Open Library
 - books, authors, publishers
- DBLP
 - computer science publications



Vocabularies

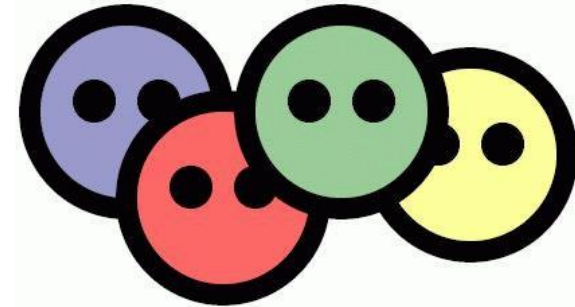
- Recap: LOD Best Practices, Principle 3:
 - Use terms from widely deployed vocabularies
- So, what are common widely deployed vocabularies?

Dublin Core



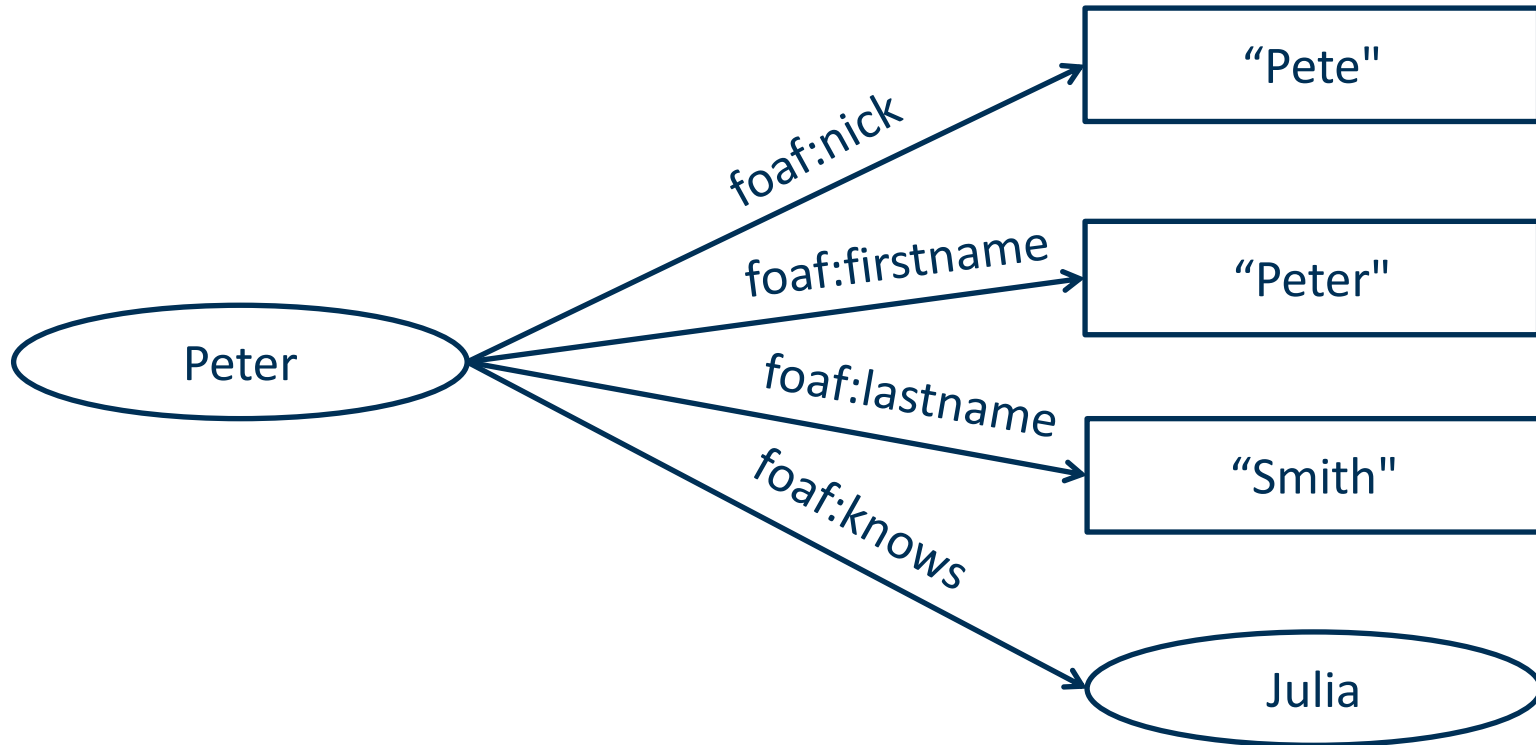
- We have already encountered this
- Usage: Metadata for resources and documents
- Namespace `http://purl.org/dc/elements/1.1/`
- Common prefix: `dc`
- Defines properties, e.g.,
 - creator
 - subject
 - date
- Resources: DCMI Type Vocabulary:
 - Text
 - Image
 - Software
 - ...

FOAF (Friend of a Friend)

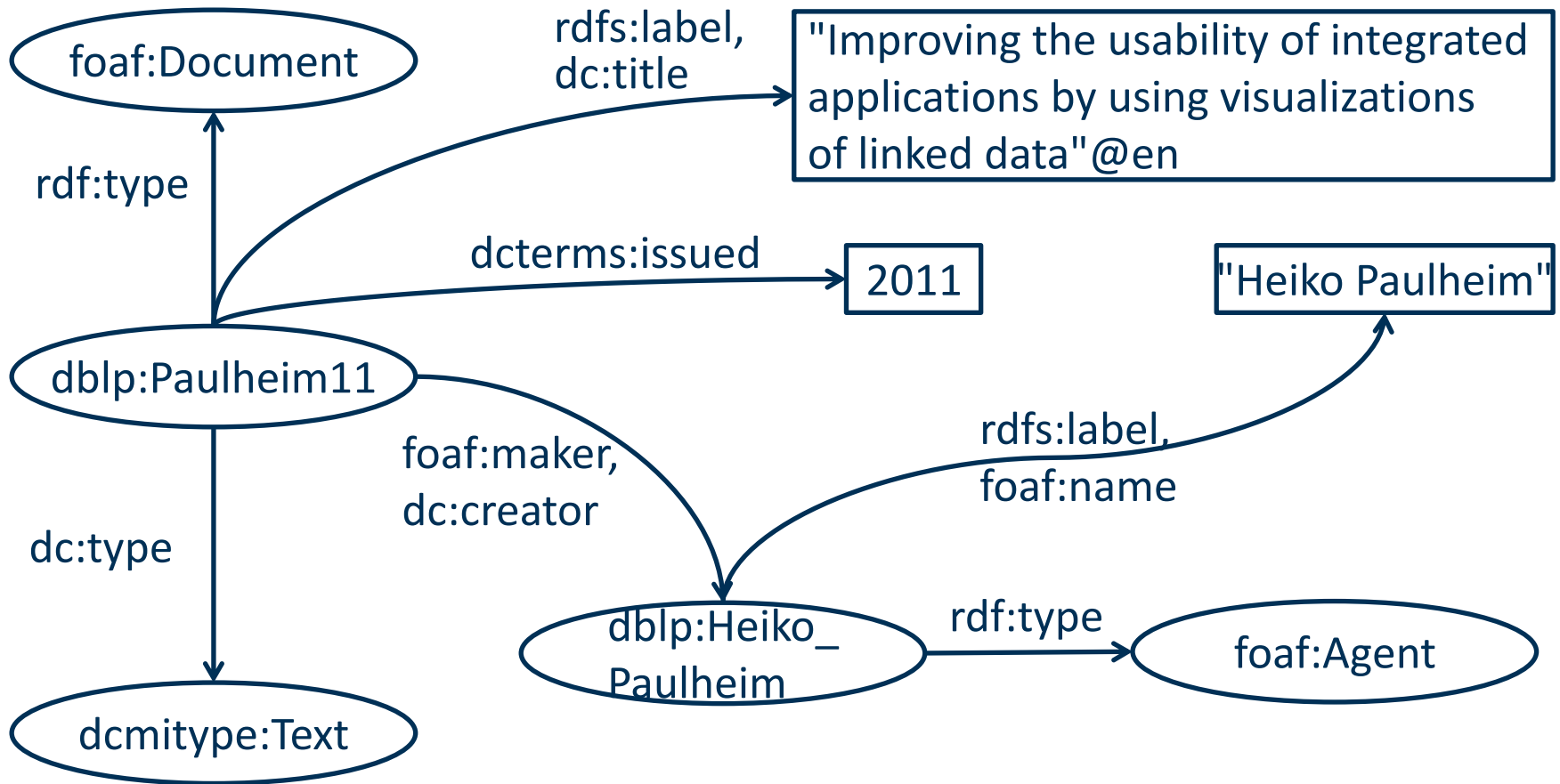


- Persons and their relations
- Created for personal home pages
 - but used widely beyond that
- Namespace `http://xmlns.com/foaf/0.1/`
- Common prefix: `foaf:`
- Important classes
 - Person
 - Group
 - Organization
 - Project
 - ...
- Important properties
 - name, firstName, lastName
 - phone, mbox, homepage
 - knows
 - currentProject, pastProject
 - ...

FOAF (Friend of a Friend)



DBLP: Combining FOAF and DC



WGS 84

- Encodes geographic data
- World Geodetic System 1984
- 3D reference model

- Namespace

`http://www.w3.org/2003/01/geo/wgs84\_pos#`

- Common prefix: `geo:`

- Classes:

- SpatialThing
- Point

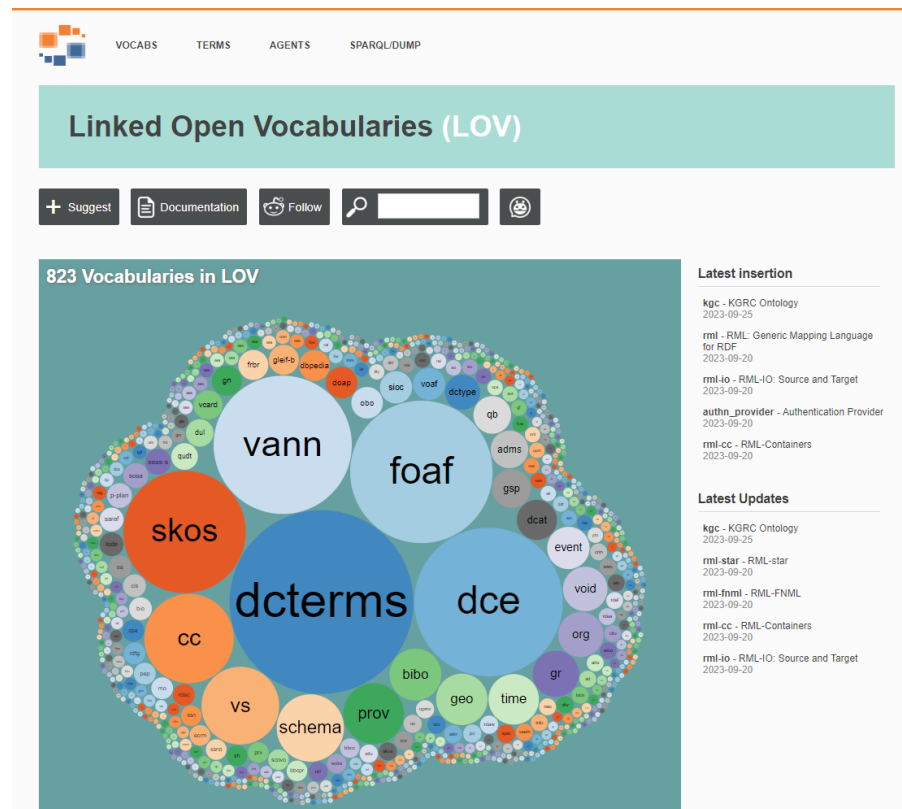
- Properties:

- latitude
- longitude
- altitude
- location



Vocabularies

- Where to search for vocabulary terms?
 - One possibility: <https://lov.linkeddata.es/dataset/lov/>



Vocabularies

- Where to search for prefix definitions?
 - One possibility: <http://prefix.cc/>



rdfs

<http://www.w3.org/2000/01/rdf-schema#> ➡ +1
-1
[Add alternative URI](#)

[ttl](#) [xml](#) [rdfa](#) [sparql](#) [txt](#) [json](#) [jsonld](#) [vann](#) | [lov](#) | [prefix.cc](#)

```
← → ↻ 🔒 w3.org/2000/01/rdf-schema#

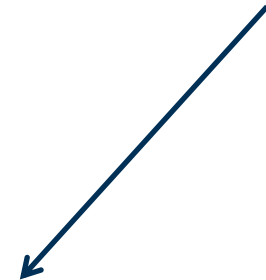
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix dc: <http://purl.org/dc/elements/1.1/> .

<http://www.w3.org/2000/01/rdf-schema#> a owl:Ontology ;
    dc:title "The RDF Schema vocabulary (RDFS)" .

rdfs:Resource a rdfs:Class ;
    rdfs:isDefinedBy <http://www.w3.org/2000/01/rdf-schema#> ;
    rdfs:label "Resource" ;
    rdfs:comment "The class resource, everything." .

rdfs:Class a rdfs:Class ;
    rdfs:isDefinedBy <http://www.w3.org/2000/01/rdf-schema#> ;
    rdfs:label "Class" ;
    rdfs:comment "The class of classes." ;
    rdfs:subClassOf rdfs:Resource .

rdfs:subClassOf a rdf:Property ;
    rdfs:isDefinedBy <http://www.w3.org/2000/01/rdf-schema#> ;
    rdfs:label "subClassOf" ;
    rdfs:comment "The subject is a subclass of a class." ;
    rdfs:range rdfs:Class ;
    rdfs:domain rdfs:Class .
```

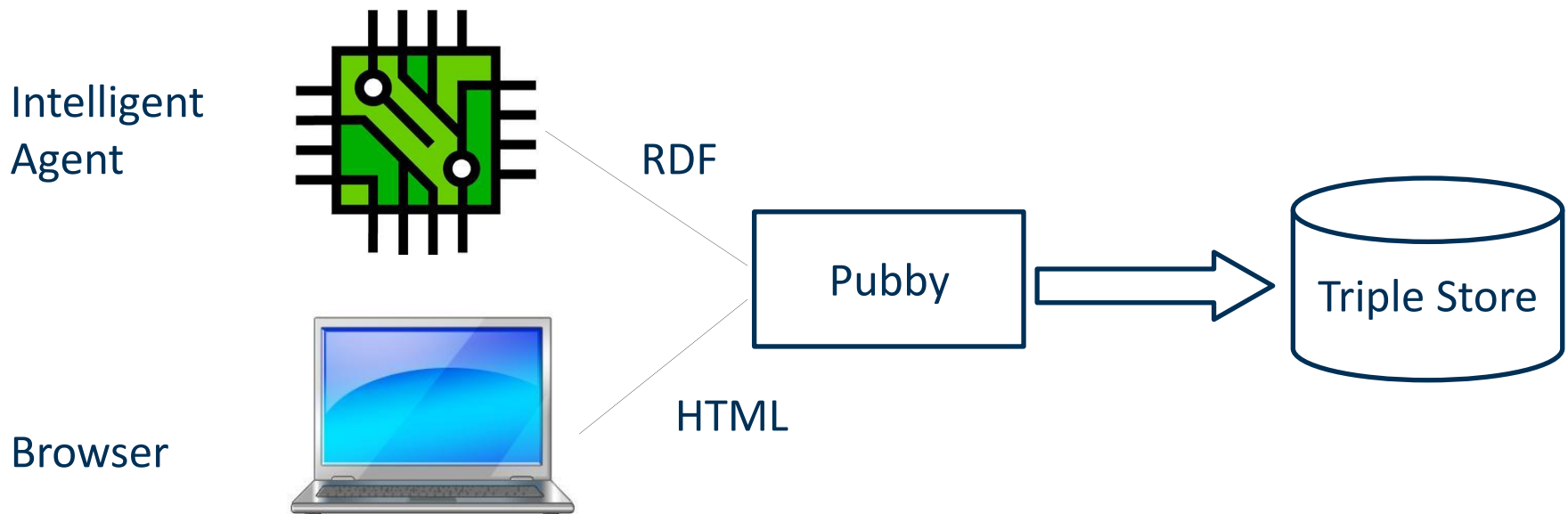


Publishing Linked Open Data

- Possible variants
 - hand coded
 - from triple stores
 - from relational databases

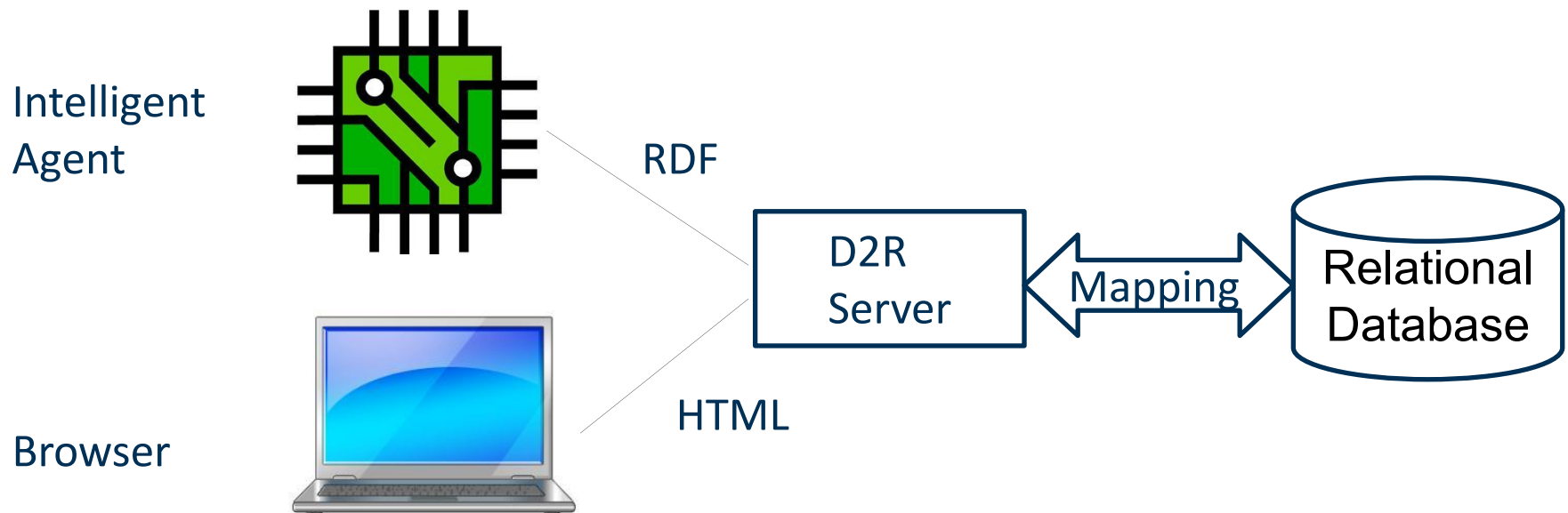
Linked Data from Triple Stores

- Triple Store: RDF storage engine
 - e.g., Virtuoso
- Pubby: Front end for triple stores
- Supports content negotiation etc.



Knowledge Graphs from Databases

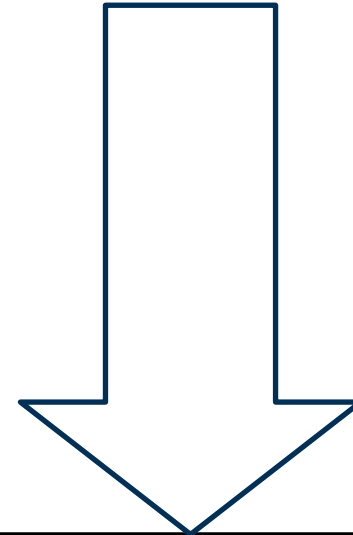
- D2R: Linked Open Data interface on relational databases
 - e.g., MySQL



Knowledge Graphs from Databases

ID (int)	name (text)	location (int)
1327890123	"Heiko"	"Mannheim"
		...

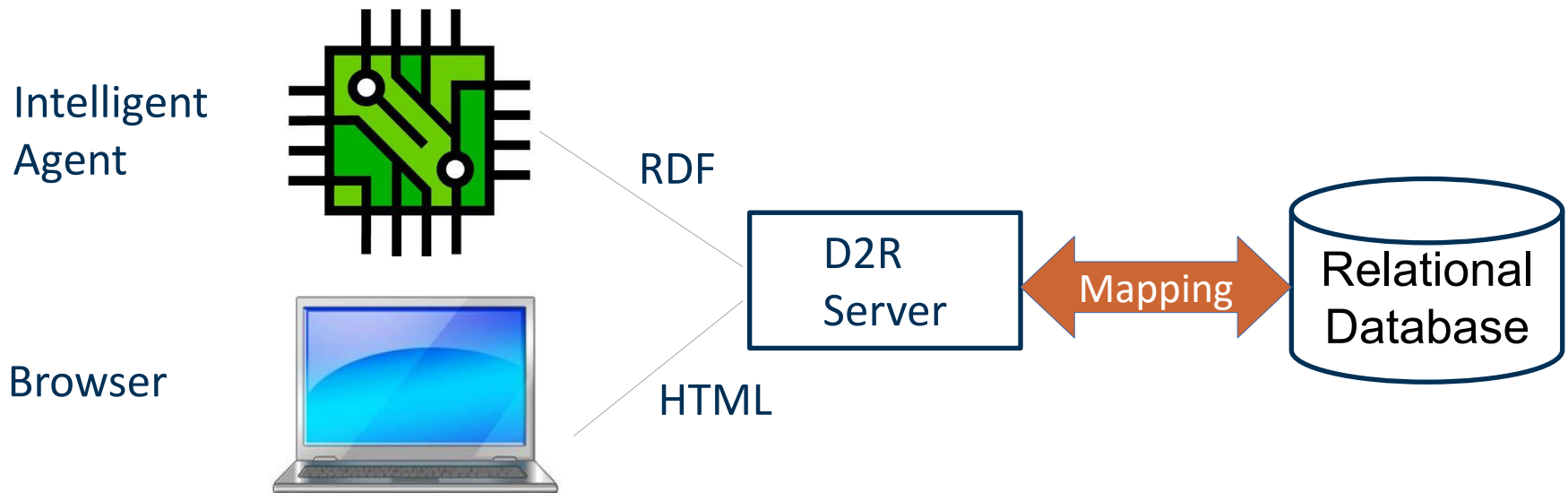
```
map:Person a d2rq:ClassMap;
  d2rq:dataStorage map:Database1.
  d2rq:class foaf:Person;
  d2rq:uriPattern "http://foo.bar/p@@Person.ID@";
.
map:personName a d2rq:PropertyBridge;
  d2rq:belongsToClassMap map:Person;
  d2rq:property foaf:name;
  d2rq:column "Person.name";
  d2rq:datatype xsd:string;
.
map:location a d2rq:PropertyBridge;
  d2rq:belongsToClassMap map:Person;
  d2rq:property foaf:basedNear;
  d2rq:column "Person.location";
  d2rq:datatype xsd:string;
.
```



```
<http://foo.bar/p1327890123> a foaf:Person .
<http://foo.bar/p1327890123> foaf:name "Heiko" .
<http://foo.bar/p1327890123> foaf:basedNear "Mannheim" .
```

Knowledge Graphs from Databases

- Note:
 - In this case, the knowledge graph does not *replicate* the data
 - It is only a “virtual” knowledge graph, providing a knowledge graph view on data from another system
 - Combining such virtual knowledge graphs can provide a *unified view* of data from different sources



Microdata and schema.org

HTML



```
<div itemscope itemtype="http://schema.org/PostalAddress">  
  <span itemprop="name">Data and Web Science Group</span>  
  <span itemprop="addressLocality">Mannheim</span>,&br/></div>
```

```
_:1 a <http://schema.org/PostalAddress> .  
_:1 <http://schema.org/name> "Data and Web Science Group" .  
_:1 <http://schema.org/addressLocality> "Mannheim" .  
_:1 <http://schema.org/postalCode> "68131" .  
_:1 <http://schema.org/adressCounty> "Germany" .
```

Microdata and schema.org

- schema.org defines (among others)
 - products
 - product offers
 - businesses and local businesses (stores, cafés, ...)
 - books, movies, records
 - events
 - recipes
 - persons
 - ...

Movie

Thing > CreativeWork > Movie

A movie.

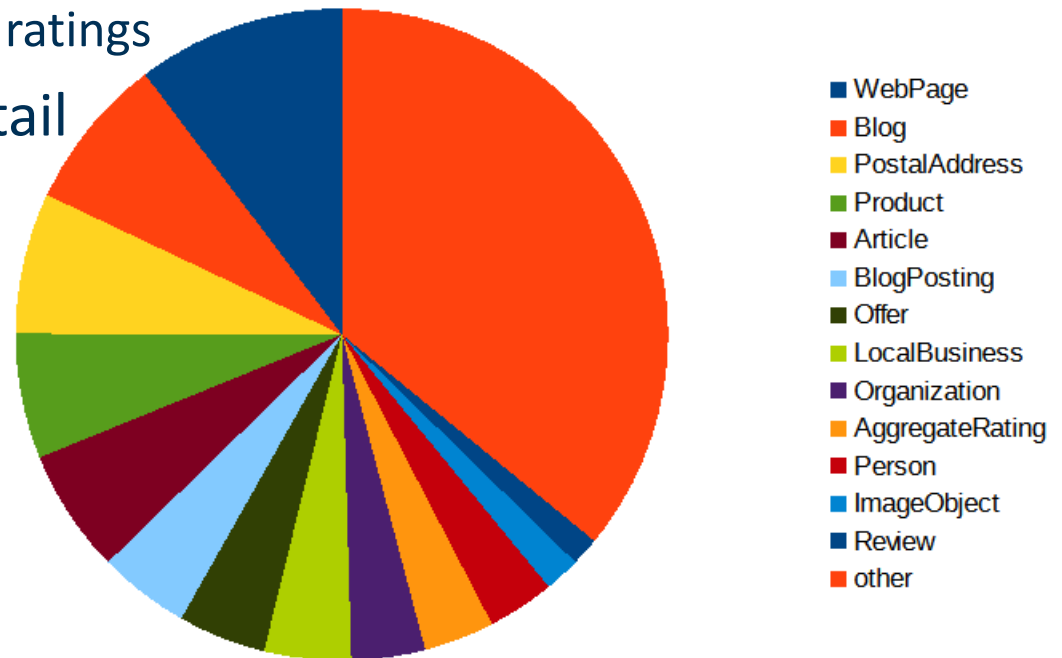
Usage: Between 10,000 and 50,000 domains

[more...]

Property	Expected Type	Description
Properties from Movie		
actor	Person	An actor, e.g. in tv, radio, movie, video games etc. Actors can be associated with individual items or with a series, episode, clip. Supersedes actors .
director	Person	A director of e.g. tv, radio, movie, video games etc. content. Directors can be associated with individual items or with a series, episode, clip. Supersedes directors .
duration	Duration	The duration of the item (movie, audio recording, event, etc.) in ISO 8601 date format .
musicBy	MusicGroup or Person	The composer of the soundtrack.
productionCompany	Organization	The production company or studio responsible for the item e.g. series, video game, episode etc.
subtitleLanguage	Text or Language	Languages in which subtitles/captions are available, in IETF BCP 47 standard format .
trailer	VideoObject	The trailer of a movie or tv/radio series, season, episode, etc.
Properties from CreativeWork		
about	Thing	The subject matter of the content.
accessibilityAPI	Text	Indicates that the resource is compatible with the referenced accessibility API (WebSchemas wiki lists possible values).
accessibilityControl	Text	Identifies input methods that are sufficient to fully control the described resource (WebSchemas wiki lists possible values).
accessibilityFeature	Text	Content features of the resource, such as accessible media, alternatives and supported enhancements for accessibility (WebSchemas wiki lists possible values).
accessibilityHazard	Text	A characteristic of the described resource that is physiologically dangerous to some users. Related to WCAG 2.0 guideline 2.3 (WebSchemas wiki lists possible values).
accountablePerson	Person	Specifies the Person that is legally accountable for the CreativeWork.
aggregateRating	AggregateRating	The overall rating, based on a collection of reviews or ratings, of the item.
alternativeHeadline	Text	A secondary title of the CreativeWork.

Deployment of schema.org

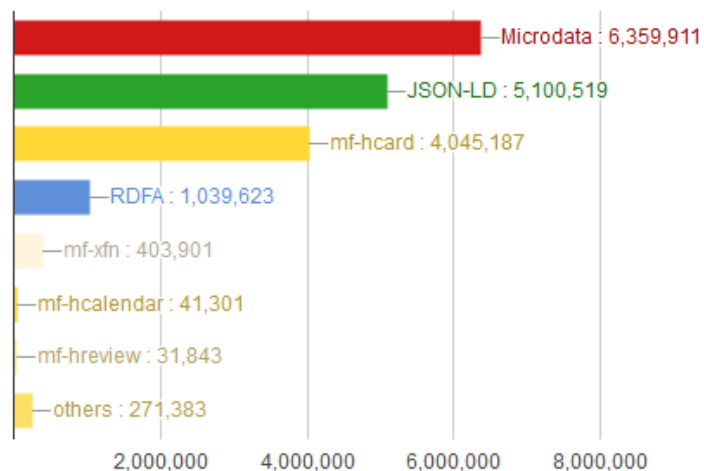
- Main topics of schema.org:
 - Meta information on web page content (web page, blog...)
 - Business data (products, offers, ...)
 - Contact data (businesses, persons, ...)
 - (Product) reviews and ratings
- ...and a massive long tail



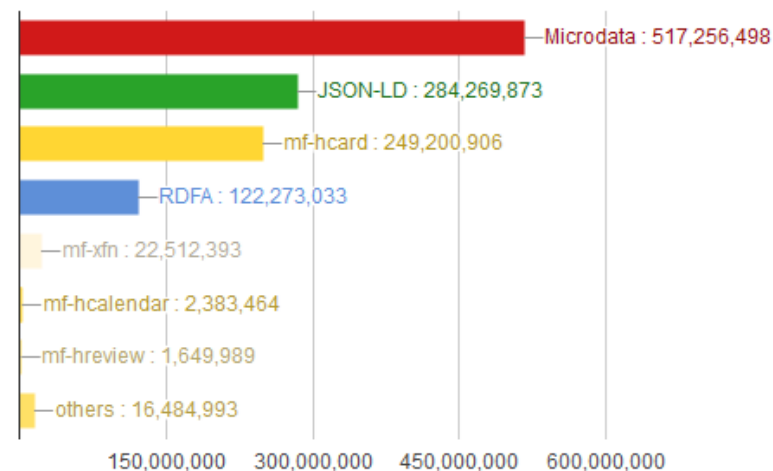
Growth of schema.org

- Note: schema.org is mainly used with Microdata
 - ...and Microdata is mainly used with schema.org

Domains with Triples



URLs with Triples



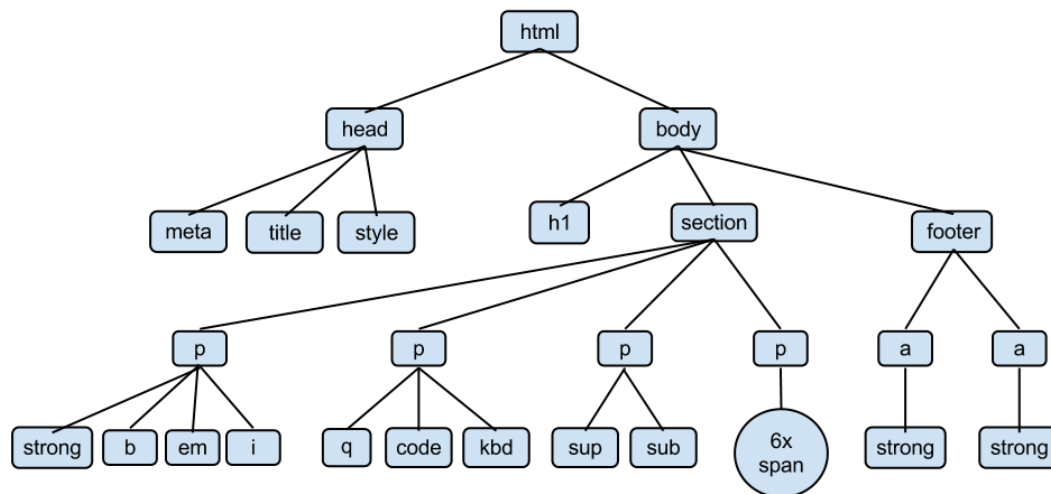
Microdata/schema.org vs. Linked Open Data

- Commonalities
 - Both encode machine-interpretable knowledge
 - Schema.org uses a standard vocabulary
 - Both can be encoded as RDF



Microdata/schema.org vs. Linked Open Data

- Differences
 - Microdata is embedded in the DOM tree
 - I.e., the resulting RDF is always a set of trees
 - Not a general directed graph
 - No cycles, no reification
 - Microdata uses only blank nodes and literals



Microdata/schema.org vs. Linked Open Data

- Linked Data Principles (Tim Berners-Lee 2006)
 - Use URIs as names for things MD2RDF creates blank nodes
 - Use HTTP URIs that can be looked up Blank nodes cannot be looked up
 - When someone looks up a HTTP URI, provide useful information using a standard HTML5+MD is a standard

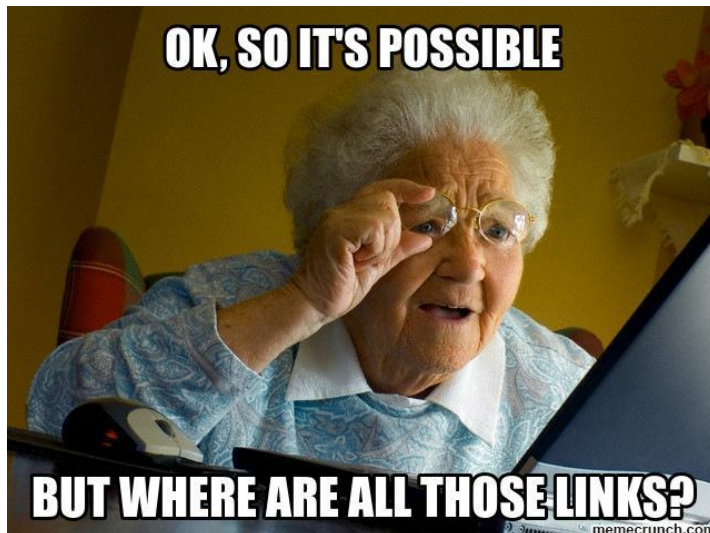
```
<div itemscope itemtype="http://schema.org/PostalAddress">  
  <span itemprop="name">Data and Web Science Group</span>  
  <span itemprop="addressLocality">Mannheim</span>,  
  <span itemprop="postalCode">68131</span>  
  <span itemprop="addressCountry">Germany</span>  
</div>
```

```
<http://foo.bar/#1> a <http://schema.org/PostalAddress> .  
<http://foo.bar/#1> <http://schema.org/name> "Data and Web Science Group".  
<http://foo.bar/#1> <http://schema.org/addressLocality> "Mannheim" .  
<http://foo.bar/#1> <http://schema.org/postalCode> "68131" .  
<http://foo.bar/#1> <http://schema.org/adressCounty> "Germany" .
```

Microdata/schema.org vs. Linked Open Data

- Linked Data Principles (TimBL 2006)
 - Use URIs as names for things
 - Use HTTP URIs that can be looked up
 - When someone looks up a HTTP URI, provide useful information using a standard
 - Include links to other URIs

This is possible with
schema.org/sameas



- Linkage within schema.org Microdata:
 - Only 0.02% of all data providers use schema.org/sameas

Microdata/schema.org vs. LOD

- Five Star Scheme (TimBL 2010)
 - * Available on the web with an open license
 - ** Available as machine-readable, structured data
 - *** as (**), using a non-proprietary format
 - **** plus: using open standards by the W3C
 - ***** plus: links to other datasets
- What's the license of web data?



Intermediate Summary

- Until today, we have dealt with the Semantic Web as a *vision*
- Today, we have seen two incarnations of that vision
 - Linked Open Data
 - schema.org/Microdata
- Both have a lot in common
- Linked Open Data:
 - A set of interconnected knowledge graphs, or a large knowledge graph
- schema.org/Microdata
 - A very large set of small knowledge graphs

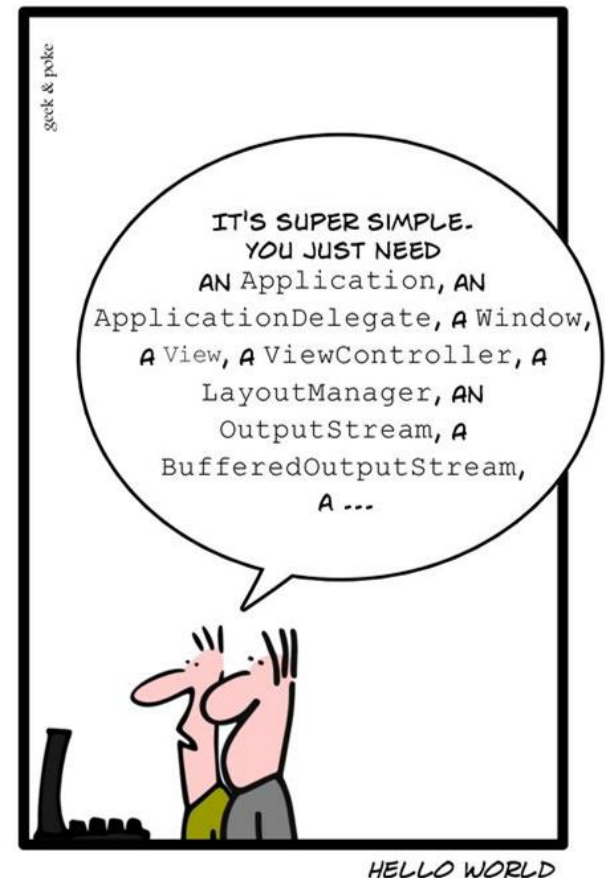
And Now for Something Completely Different



Programming with Knowledge Graphs

- Let's start with a simple application
 - a Hello World application for reading data from a knowledge graph

SIMPLY EXPLAINED



Using only Plain Java

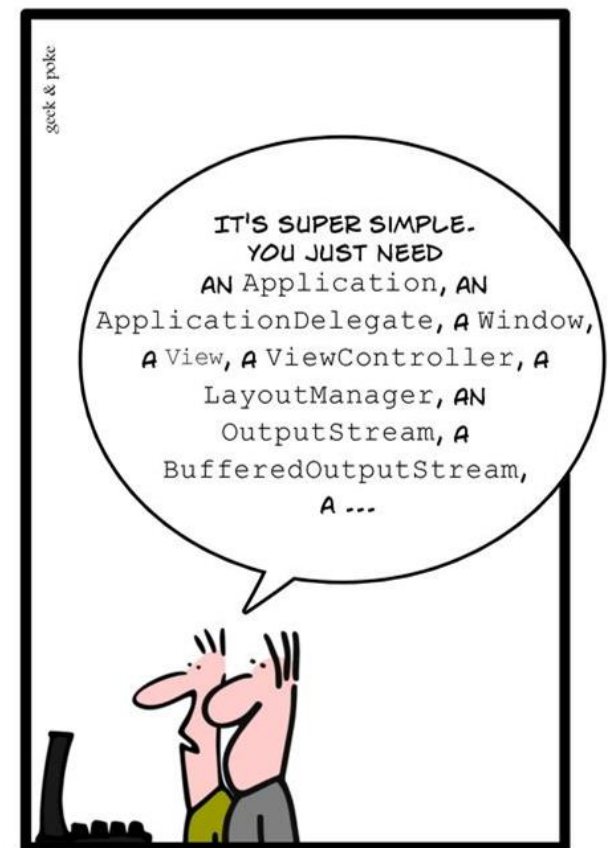
```
URL url = new URL("http://dbpedia.org/resource/Mannheim");
URLConnection conn = url.openConnection();
conn.addRequestProperty("Accept", "text/rdf+n3");
BufferedReader BR = new BufferedReader(
    new
    InputStreamReader(conn.getInputStream())
);

while (BR.ready()) {
    String triple = BR.readLine();
    StringTokenizer tokenizer = new StringTokenizer(triple, " ");
    String subject = tokenizer.nextToken();
    String predicate = tokenizer.nextToken();
    String object = tokenizer.nextToken();
    ...
}
```

Using only Plain Java

- Let's start with a simple application
 - a Hello World application for reading data from a knowledge graph
- Using plain Java is possible
 - but not very comfortable
 - there are more sophisticated frameworks

SIMPLY EXPLAINED



HELLO WORLD

Programming with Jena



- Jena is a well-known Semantic Web programming framework
- started in 2000 at HP Labs
- Apache open source project since 2010
- Central concepts
 - Models (`class Model`) are RDF graphs
 - Resources (`class Resource`) are resources in RDF graphs
- Special features
 - Database connectors for persistence
 - Support for reasoning
 - Rule engines
 - Support for SPARQL (see next lecture)

Programming with Jena

- Reading a model from a derefencable URI

```
model.read("http://dbpedia.org/resource/Mannheim");
```

- Navigating within a model

```
Resource mannheim =  
    model.getResource("http://dbpedia.org/resource/  
                      Mannheim");  
  
Resource countryOfMannheim =  
    model.getProperty(  
        "http://dbpedia.org/ontology/country"  
    ).getResource();
```

Programming with Jena

- Working with literals

```
Literal lit = mannheim.getProperty(  
    "http://www.w3.org/2000/01/rdf-schema#label"  
).getLiteral();  
  
lit.getString();  
lit.getLanguage();  
lit.getDatatype();
```

Programming with Jena

- Working with multi-valued relations

creates an iterator over all
Triples with the subject node
and the given predicate

```
- StmtIterator iter = mannheim.getProperty(  
    "http://www.w3.org/2000/01/rdf-schema#label");  
- while(iter.hasNext()) {  
    Statement s = iter.next();  
    RDFNode node = s.getObject();  
    if(node.isLiteral())  
        System.out.println(node.asLiteral().getString());  
}
```

Iterators in Jena

- Jena uses the iterator pattern quite frequently

- e.g.:

```
StmtIterator iter = mannheim.getProperty(  
    "http://www.w3.org/2000/01/rdf-schema#label");
```

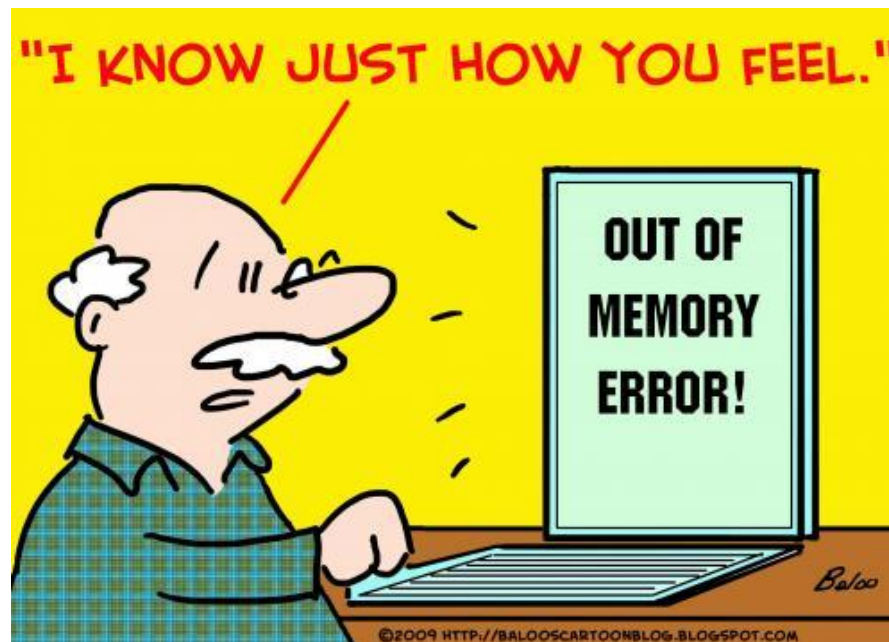
- But there is no such thing as

```
Collection<Statement> triples = mannheim.getProperty(  
    "http://www.w3.org/2000/01/rdf-schema#label");
```

- Why?

Iterators in Jena

- Knowledge graphs can be very large
- E.g., reading all triples from DBpedia
 - Stored in List<Statement> would kill the main memory
 - Iterators allow a more efficient memory use



Reasoning with Jena

- Recap: we can derive information from a schema (T-Box) and data (A-box)

`:knows rdfs:domain :Person .`

`:knows rdfs:range :Person .`

`:Peter :knows :Tom .`

`→ :Peter a :Person .`

`→ :Tom a :Person .`

- Jena also supports reasoning

Reasoning with Jena

- Given: a schema and some data

```
Model schemaModel = ModelFactory.createDefaultModel();  
InputStream IS = new  
    FileInputStream("data/example_schema.rdf");  
schemaModel.read(IS);
```

```
Model dataModel = ModelFactory.createDefaultModel();  
IS = new FileInputStream("data/example_data.rdf");  
dataModel.read(IS);
```

```
Model reasoningModel =  
    ModelFactory.createRDFSModel(schemaModel, dataModel);
```

- Now, reasoningModel contains all derived facts

UNIVERSITY
OF MANNHEIM
Data and Web Science Group

- ```
while(it.hasNext()) {
 Statement s = it.next();
 System.out.println(s);
}
```

- [illegible]

# Programming with RDFLib (Python)

- RDFLib is a Python library for working with RDF
- Initial release 4 June, 2002 by Daniel Krech
  - Now being developed by the community at github:  
<https://github.com/RDFLib/rdflib/>
- It contains parsers and serializers for
  - RDF/XML, N3, NTriples, N-Quads, Turtle, TriX, RDFa and Microdata
- Graph interface which can be backed by store implementations
  - Memory storage
  - Persistent storage on top of the Berkeley DB
- Reasoning possible (<https://github.com/RDFLib/OWL-RL> )
- SPARQL 1.1 implementation (see next lecture)

# Programming with RDFLib (Python)

- primary interface is a Graph
    - represented as a set of 3-item triples
- ```
[  
    (subject1, predicate1, object1),  
    (subject2, predicate2, object2),  
    ...  
    (subjectN, predicateN, objectN)  
]
```

Programming with RDFLib (Python)

- Reading a model from a derefencable URI

```
import rdflib
g=rdflib.Graph()
g.load('http://dbpedia.org/resource/Mannheim')
```

- Print out all RDF triples

```
for s,p,o in g:
    print(s,p,o)
```

- Navigating within a graph

```
print(g.value(
    URIRef("http://dbpedia.org/resource/Mannheim"),
    URIRef("http://dbpedia.org/ontology/country")
))
```

Programming with RDFLib (Python)

- Most often reduced to basic triple matching
- `Graph.triples(subject, predicate, object)`
 - each of them can be `None` (similar to null in Java)

```
for s,p,o in g.triples( (None, RDF.type, FOAF.Person) ):
    print("%s is a person"%s)
```
- Special functions for returning only specific parts
 - `Graph.subjects(predicate, object)` – returns only subjects
 - `Graph.predicate(subject, object)`
 - `Graph.objects(subject, predicate)`
 - `Graph.subject_objects(predicate)`
 - `Graph.subject_predicates(object)`
 - `Graph.predicate_objects(subject)`
 - `Graph.value(subject, predicate)`

Programming with RDFLib (Python)

- Create URIs

```
mannheim = URIRef('http://example.com/Mannheim')
```

- Create literals

```
mannheim_literal = Literal("Mannheim")
```

- Add triples to graph

```
g.add( (mannheim, RDFS.label, mannheim_literal) )  
g.add( (mannheim, RDFS.label, Literal("Mannheim")) )
```

- Serialize graph

```
print( g.serialize(format='n3') )
```

Wrap-Up

- Today, we have seen
 - Two incarnations of knowledge graphs as publicly available data
 - i.e., Linked Open Data
 - and Microdata/schema.org
- ...and we have learned how to write programs consuming data in knowledge graphs
 - Jena & RDFlib programming frameworks
 - Loading RDF from files and from URLs
 - Performing reasoning

Questions?

